

Experiment 1

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1 Aim

In this experiment we aim to study the Zener diode as a voltage regulator and the use of IC 7805 as a voltage stabilizer.

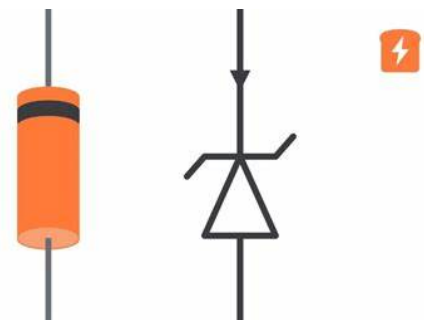
2 Experimental Setup

Zener Diode

A Zener diode is a device which operates in reverse bias mode. When the reverse voltage exceeds the breakdown voltage (V_z), the diode starts to conduct, maintaining a stable voltage across the load. This makes it suitable for voltage regulation applications.



(a) Zener Diode appearance

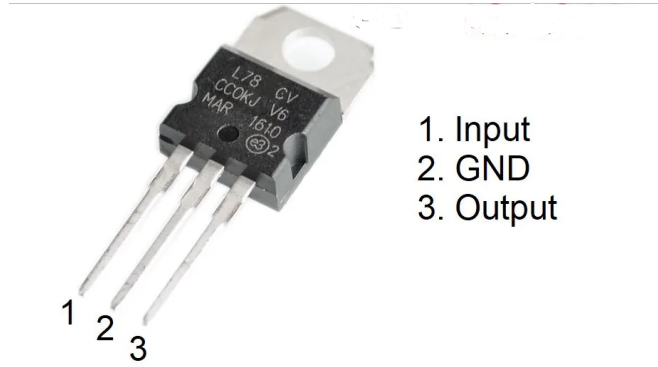


(b) Polarity of a zener diode

IC 7805 Voltage Regulator

IC 7805 is a linear voltage regulator that provides a stable 5V output for a range of input voltages (7.2V - 35V). It can supply up to 1A of current.

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3 Theory

Load Regulation

Load regulation refers to the ability of a voltage regulator to maintain a constant output voltage despite variations in the load resistance. It ensures that the output voltage remains stable even when the load varies.

- Mathematically, load regulation is given by:

$$\text{Load Regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$$

where:

V_{NL} is the output voltage at no load.

V_{FL} is the output voltage at full load.

- Experimental Method:
 - In this experiment, load regulation is tested by varying the load resistance (R_L) while keeping the input voltage (V_{in}) constant.
 - The output voltage (V_{out}) is measured for different values of R_L .
 - The change in output voltage is recorded to determine how well the regulator maintains stability under varying loads.

Line Regulation

Line regulation measures how well a voltage regulator maintains the output voltage when the input voltage varies. It ensures that changes in input voltage do not significantly affect the output voltage.

- Mathematically, it is given by:

$$\text{Line Regulation} = \frac{\Delta V_{out}}{\Delta V_{in}}$$

where:

ΔV_{out} is the change in output voltage.

ΔV_{in} is the change in input voltage.

- Experimental Method:
 - Line regulation is tested by varying the input voltage (V_{in}) while keeping the load resistance (R_L) constant.
 - The output voltage (V_{out}) is measured for different values of R_L .
 - The change in output voltage is analyzed to determine the effectiveness of the voltage regulator.

Breakdown Voltage

Breakdown voltage is the minimum reverse voltage at which a diode, specifically a Zener diode, begins to conduct in reverse and maintains a stable voltage.

4 Data Analysis

Part-A: Zener Diode

Line Regulation

In *line regulation*, the series limiting resistance ($R_s = 2.2 \text{ k}\Omega$) and load R_L (1.006 k Ω potentiometer) is kept fixed. The input voltage V_{in} is varied and V_{out} is measured (Table 1). In this setup we measure the currents i_s and i_z with ammeters.

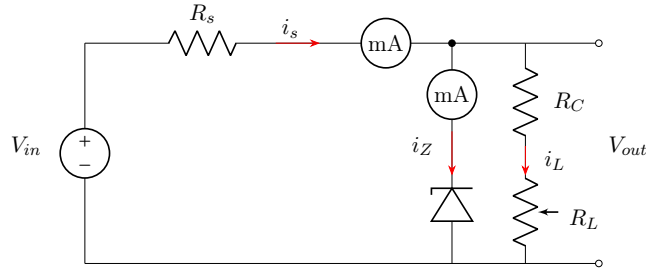


Figure 2: The circuit diagram of the setup. R_c is 2.2 k Ω and V_{out} is measured with a multimeter.

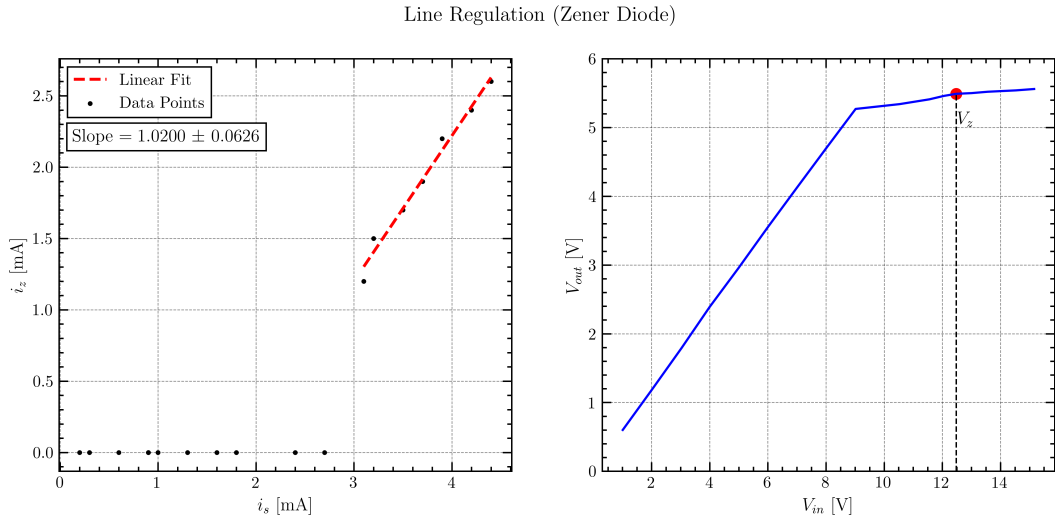


Figure 3: Left: A plot of i_s versus the Zener current i_z is shown. The positive slope of 1.0200 ± 0.0626 confirms the successful verification of $\delta i_s = \delta i_z$. **Right:** A graph of V_{out} versus V_{in} is presented, where the breakdown voltage is identified as the point beyond which V_{out} becomes approximately constant.

The positive slope in Fig 3 (Left) of 1.0200 ± 0.0626 successfully verifies the direct proportionality between δi_s and δi_z . From Fig 3 (Right) we obtain the $V_{in} = 12.48 \text{ V}$. Therefore,

$$V_z = V_{in} - i_s R_s = 12.48 - 3.2 \times 2.2 = 5.44$$

Hence the breakdown voltage is calculated as 5.44 V. Thus when load is fixed and input varies, i_z and i_s change in such a way as to maintain i_L and hence V_{out} constant. Any change in V_i appears across R_s .

Load Regulation (without R_c)

In *load regulation* we study the variation of i_z with i_L , by varying R_L and holding V_{in} constant. If $\Delta V_{diode} \geq V_z$, then the voltage across zener remains constant and i_s becomes independent of load. Thus

in this case we expect to see,

$$\delta i_z = -\delta i_L$$

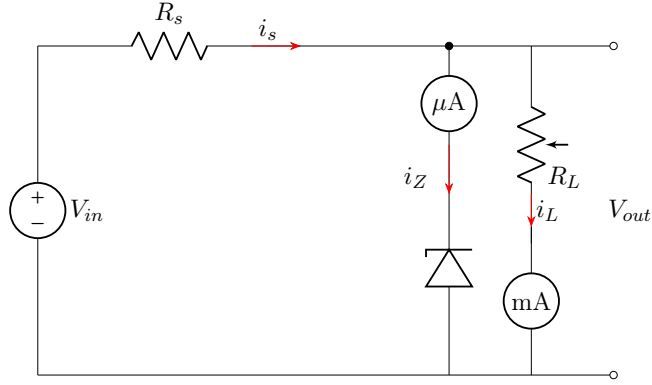


Figure 4: The circuit diagram of this setup.

Load Regulation (Zener Diode ; $R_C = 0$)

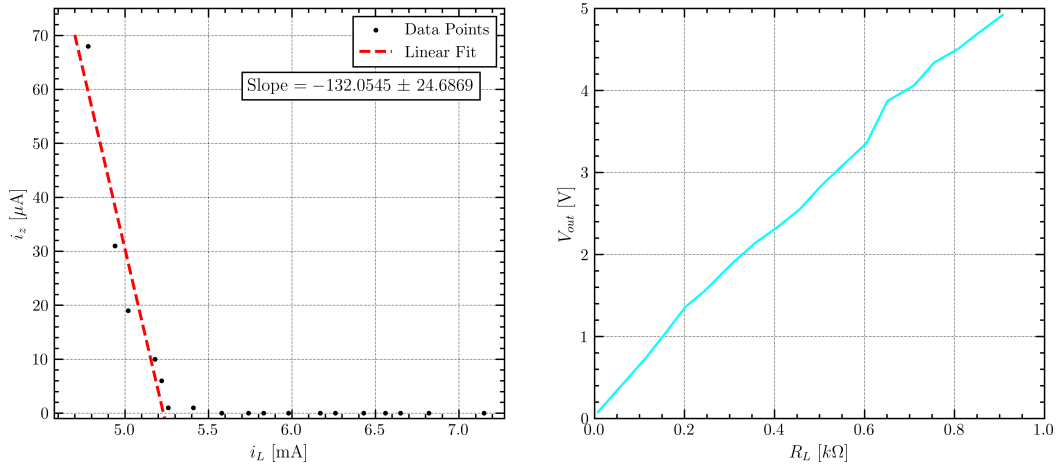


Figure 5: Left: This plot demonstrates that the Zener diode does not reach the breakdown region. Although $\delta i_z \propto -\delta i_L$, the two quantities are not equal. **Right:** This plot shows that V_{out} is never constant, even when R_L is set to its maximum value.

In this setup (Table 2), we find that V_{out} never reaches V_z despite increasing R_L to the maximum (Fig 5). This can be explained as follows:

$$\Delta V_{diode} = V_{in} - R_s \frac{V_{in}}{R_s + R_L} = V_{in} \frac{R_L}{R_L + 2.2} \leq \frac{15.59 \times 1}{1 + 2.2} = 4.87 \text{ V} < V_z$$

Hence we were not be able to find the zener breakdown voltage from this setup.

Load Regulation (with R_c)

The negative slope of -0.9596 ± 0.0573 in Fig 7, successfully verifies the inverse proportionality $\delta i_z = -\delta i_L$ (Table 3). In the next graph, we calculate the average output voltage to be $5.51 \pm 0.02 \text{ V}$. Ideally this should be *equal* to the breakdown voltage V_z of the diode, but the reason for not being may be attributed to the additional resistance of wires which creates a greater potential difference.

Thus, zener current inversely changes with change in load current due to change in R_L , but output remains constant

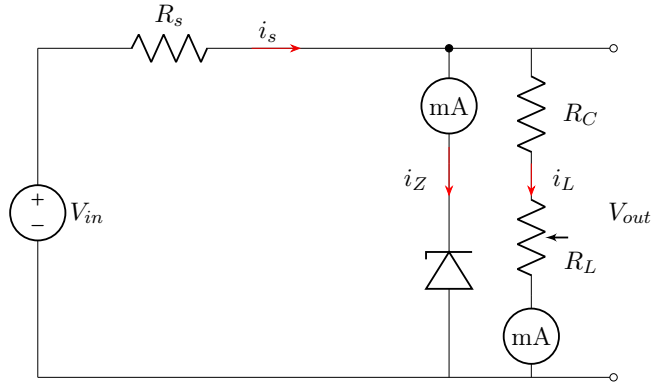


Figure 6: The circuit diagram of this setup.

Load Regulation (Zener Diode ; $R_C = 2.2k\Omega$)

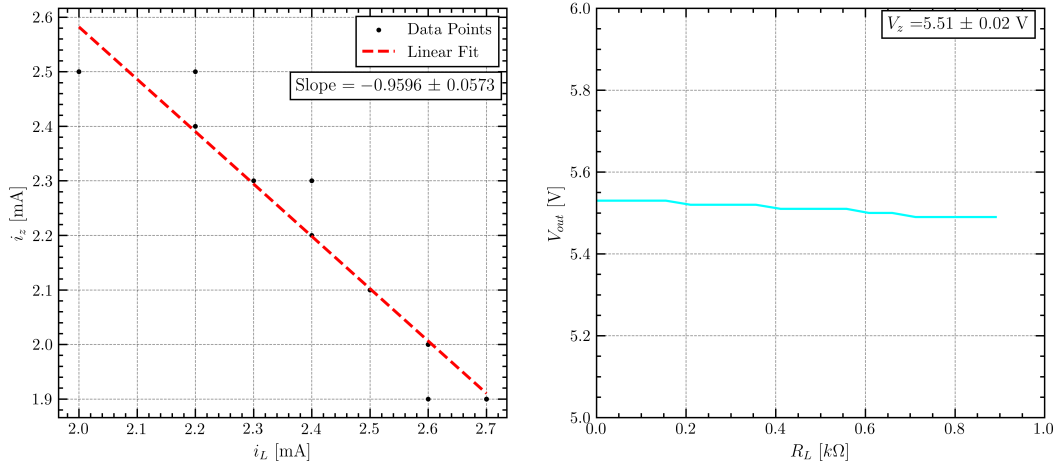


Figure 7: Left: A plot of the Zener current (i_Z) versus the load current (i_L) is shown. A linear fit with a slope of -0.9596 ± 0.0573 confirms the inverse proportionality between the change in two currents, $\delta i_Z \approx -\delta i_L$. **Right:** Here, V_{out} is observed to remain approximately constant, indicating that the diode operates in the breakdown region across the entire range of R_L .

Part-B: IC 7805 Voltage Regulator

Line Regulation

In this setup, we vary V_{in} (Table 4) keeping $(R_C + R_L)$ constant and measure V_{out} and i_L .

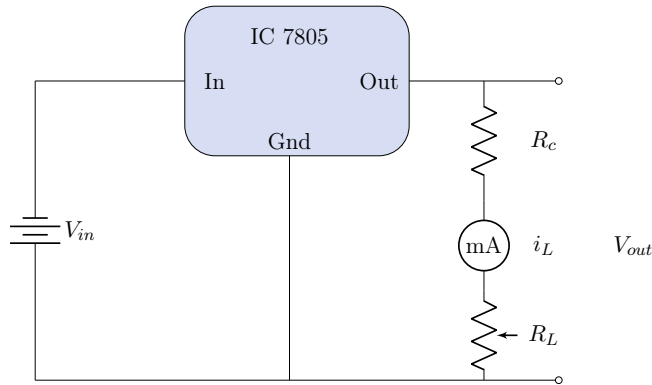


Figure 8: Circuit diagram for this part of the experiment

From the above graph (Fig 9) we can identify the minimum input voltage, $V_{min} = 6.12$ V at which the output V_{out} becomes constant at 3.85 V

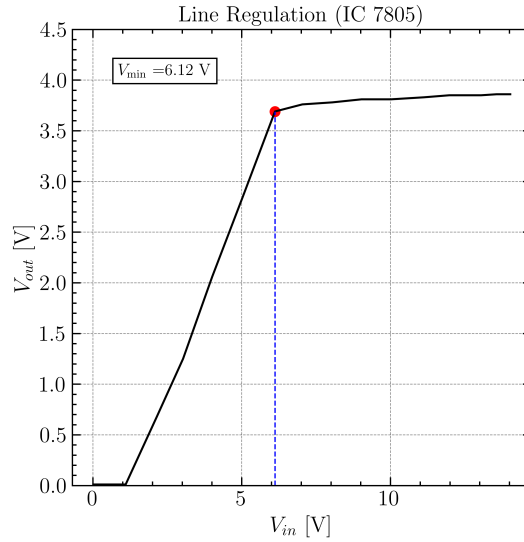


Figure 9: The graph of line regulation for voltage regulator

Load Regulation

In this setup (Table 5) the input voltage, V_{in} remains constant and the load resistance, R_L is varied. However the output voltage was observed to be the same irrespective of load.

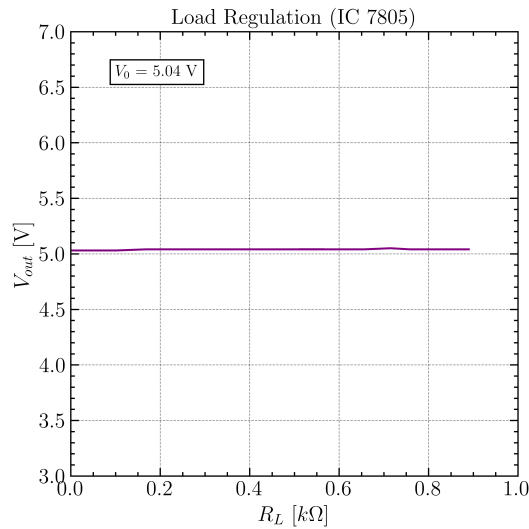


Figure 10: Load regulation for voltage regulator

From the above graph (Fig 9) we find V_{out} to be constant at 5.04 V.

5 Sources of Error

- Voltage drop across Zener Diode (V_z) is not equal to output voltage measured (V_{out}) as we expect ideally. This can be due to improper calibration of multimeter and resistance in wires.
- The Zener diode and voltage regulator ICs are temperature-dependent. Heating due to prolonged operation may shift the breakdown voltage and regulation behaviour.
- Loose connections or improper placement of wires on the breadboard could introduce additional resistance and voltage drops.
- The output of the given voltage source was not steady and was fluctuating.

6 Conclusion

We studied the line and load regulation of the Zener diode. From the line regulation, the breakdown voltage was observed to be $5.54 \pm 0.02\text{V}$ despite variations in input voltage or load resistance. We found out that the voltage across the diode remains almost constant for increasing input voltage once breakdown voltage is reached. In the next part of experiment we implemented IC 7805 as a voltage regulator. From the line regulation, we found out that the minimum voltage required for the IC to achieve a stable voltage was 6.12 V. The stable voltage for the IC was found to be 5.04 V as expected. Thus the overall experiment we can conclude that the IC 7805 is a better voltage regulator than the Zener diode.

References

- *B. Ghosh*, Fundamental Principles of Electronics 2021. ISBN: 9788187134398

Appendix

Sl. No	R_L (k Ω)	V_i (V)	i_s (mA)	i_z (mA)	V_0 (V)
1	1.006	1.01	0.2	0.0	0.60
2	1.006	2.02	0.3	0.0	1.19
3	1.006	3.03	0.6	0.0	1.79
4	1.006	4.04	0.8	0.0	2.39
5	1.006	5.00	0.9	0.0	2.96
6	1.006	6.00	1.2	0.0	3.55
7	1.006	7.01	1.3	0.0	4.13
8	1.006	8.06	1.6	0.0	4.68
9	1.006	9.02	1.8	0.1	5.06
10	1.006	10.01	2.2	0.5	5.27
11	1.006	10.53	2.4	0.7	5.34
12	1.006	11.20	2.7	0.9	5.41
13	1.006	11.49	2.8	1.0	5.43
14	1.006	12.06	3.1	1.2	5.46
15	1.006	12.48	3.2	1.5	5.48
16	1.006	13.00	3.5	1.7	5.50
17	1.006	13.56	3.7	2.0	5.52
18	1.006	14.04	3.9	2.2	5.54
19	1.006	14.53	4.2	2.4	5.54
20	1.006	15.16	4.4	2.6	5.56

Table 1: Data for Line regulation of Zener Diode

Sl. No	V_i (V)	R_L (k Ω)	i_L (mA)	i_z (μ A)	V_0 (V)
1	15.59	0.008	7.15	0	0.08
2	15.59	0.113	6.82	0	0.73
3	15.59	0.159	6.65	0	1.05
4	15.59	0.202	6.56	0	1.29
5	15.59	0.246	6.43	0	1.56
6	15.59	0.306	6.28	0	1.89
7	15.59	0.357	6.17	0	2.14
8	15.59	0.400	6.08	0	2.33
9	15.59	0.446	5.98	0	2.55
10	15.59	0.506	5.83	0	2.85
11	15.59	0.558	5.74	0	3.07
12	15.59	0.605	5.45	1	3.69
13	15.59	0.651	5.36	2	3.87
14	15.59	0.710	5.22	6	4.16
15	15.59	0.755	4.83	10	4.34
16	15.59	0.809	5.02	19	4.51
17	15.59	0.846	4.94	31	4.67
18	15.59	0.907	4.78	68	4.92

Table 2: Load Regulation of Zener diode without current limiting resistor (R_C)

Sl. No	V_i (V)	R_L (k Ω)	i_z (mA)	i_L (mA)	V_0 (V)
1	15.55	0.002	2.0	2.5	5.53
2	15.55	0.031	2.0	2.5	5.53
3	15.55	0.058	2.2	2.5	5.53
4	15.55	0.106	2.2	2.4	5.53
5	15.55	0.155	2.2	2.4	5.53
6	15.55	0.210	2.3	2.3	5.53
7	15.55	0.255	2.4	2.3	5.52
8	15.55	0.308	2.4	2.2	5.52
9	15.55	0.356	2.4	2.2	5.52
10	15.55	0.411	2.5	2.1	5.51
11	15.55	0.456	2.5	2.1	5.51
12	15.55	0.502	2.5	2.1	5.51
13	15.55	0.558	2.6	2.0	5.50
14	15.55	0.608	2.6	2.0	5.50
15	15.55	0.660	2.6	2.0	5.50
16	15.55	0.712	2.6	2.0	5.50
17	15.55	0.754	2.7	1.9	5.49
18	15.55	0.802	2.7	1.9	5.49

Table 3: Load Regulation of Zener diode with current limiting resistor (R_C)

Sl No.	$R_c + R_L$ (k Ω)	V_i (V)	i_L (mA)	V_0 (V)
1	3.18	0.01	0.00	0.01
2	3.18	1.10	0.00	0.01
3	3.18	2.15	0.21	0.68
4	3.18	3.03	0.38	1.25
5	3.18	4.01	0.63	2.06
6	3.18	5.04	0.87	2.85
7	3.18	6.12	1.13	3.69
8	3.18	7.04	1.15	3.76
9	3.18	8.05	1.16	3.78
10	3.18	9.03	1.16	3.81
11	3.18	10.02	1.16	3.81
12	3.18	11.13	1.17	3.83
13	3.18	11.98	1.17	3.85
14	3.18	12.52	1.17	3.85
15	3.18	13.05	1.18	3.85
16	3.18	13.57	1.18	3.86
17	3.18	14.03	1.18	3.86

Table 4: Data for Line Regulation of IC 7805

Sl. No.	V_i (V)	R_L (k Ω)	i_L (mA)	V_o (V)
1	15	0.001	2.25	5.03
2	15	0.050	2.2	5.03
3	15	0.102	2.15	5.03
4	15	0.168	2	5.04
5	15	0.211	2.06	5.04
6	15	0.243	2.03	5.04
7	15	0.303	1.98	5.04
8	15	0.355	1.94	5.04
9	15	0.413	1.9	5.04
10	15	0.451	1.87	5.04
11	15	0.507	1.84	5.04
12	15	0.566	1.8	5.04
13	15	0.511	1.78	5.04
14	15	0.656	1.75	5.04
15	15	0.715	1.73	5.05
16	15	0.758	1.7	5.04
17	15	0.815	1.67	5.04
18	15	0.866	1.65	5.04
19	15	0.890	1.64	5.04

Table 5: Data for Load Regulation of IC 7805